

7. [Maximum mark: 7]

Two machines are used in a factory to manufacture semiconductors. Machine A manufactures defective semiconductors 10% of the time, while machine B manufactures defective semiconductors 5% of the time.

A randomly selected machine manufactures ten semiconductors.

Both machines are equally likely to be selected.

(a) Find the probability that exactly two semiconductors are defective. [4]

(b) Given that exactly two semiconductors are defective, find the probability that they were manufactured by machine A. [3]

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